

Multi-Objective Grey Wolf Optimization of Temporal Convolutional Networks for Multi-Step ECG Forecasting

Ibrahim Hanafy

Nile University Research Internship Program

Time Series Forecasting Group

Giza, Egypt

s-ibrahim.hanafy@zewailcity.edu.eg

Abstract—Forecasting the electrocardiogram (ECG) waveform, rather than classifying beats after the fact, turns cardiac monitoring into an anticipatory task, but it demands models that are accurate over a multi-sample horizon and cheap enough to retrain for each patient. Temporal Convolutional Networks (TCN) are studied here for that problem on the MIT-BIH Arrhythmia Database under a strictly controlled protocol: identical splits, identical training budget, and one model per record. Three findings are reported. First, receptive-field coverage dominates width: replacing the single-block, 256-filter TCN of the reference benchmark [3] with an initial proposed architecture of three blocks and 64 filters raises the receptive field from 5 to 29 samples, cuts parameters by $3.9\times$, and reduces RMSE by 5.7–14.1% across horizons $H \in \{1, 10, 15, 20\}$ while training 17–25% faster. Second, metaheuristic tuning helps, but single-objective search buys accuracy with size: the Grey Wolf Optimizer selects a 1.46M-parameter model for a 2.4% RMSE gain at $3.7\times$ the training time, whereas its multi-objective variant, optimizing RMSE and training time jointly, recovers statistically indistinguishable accuracy from a Pareto-optimal configuration that is $5.4\times$ smaller and $2.2\times$ faster to train. Third, at the single-step horizon reported in [3], the tuned TCN attains a mean RMSE of 0.00499 on the same 21 records against 0.0082 for the best of the twelve architectures benchmarked there, a 39.1% reduction, winning on 20 of 21 records (Wilcoxon $p=2.4\times 10^{-5}$). Scaled to all 48 MIT-BIH records the same configuration yields $R^2=0.9970$ at $H=1$ and $R^2=0.8865$ at $H=10$.

Index Terms—Temporal convolutional network, ECG forecasting, multi-objective optimization, Grey Wolf Optimizer, hyperparameter optimization, MIT-BIH

I. INTRODUCTION

Most deep learning work on the electrocardiogram is retrospective: a beat is recorded, then labelled. Forecasting inverts that order. If the next tens of milliseconds of the waveform can be predicted from the recent past, a monitor can flag a divergence between prediction and observation as it develops, rather than after a complete abnormal beat has been recorded [1]. The horizon matters clinically: predicting H samples ahead at 360 Hz gives a window of $H/360$ seconds, and a multi-output model that emits all H values in one forward pass avoids the error compounding that afflicts recursive schemes [2].

Temporal Convolutional Networks are an attractive backbone. Causal dilated convolutions give an exponentially grow-

ing receptive field at fixed depth, inference latency is deterministic, and the sequence is processed in parallel [4]. Twelve architectures spanning classical, recurrent, convolutional, and hybrid designs were benchmarked on 21 MIT-BIH records treated as chaotic time series in [3], where TCN–RNN hybrids led the field. That study fixes both the protocol and the reference numbers used throughout this paper.

Two questions are left open. The first is architectural. The TCN configuration used in [3] is wide and shallow, one residual block with 256 filters, which yields a receptive field of 5 samples against a 10-sample input window: the convolutional stack cannot see the whole input, and the dense head must compensate. Whether depth, and hence receptive-field coverage, matters more than width is not tested there. The second is methodological. Hyperparameters were selected there by grid search, which is exhaustive but does not scale to continuous parameters such as the learning rate. Metaheuristic search does, and the Grey Wolf Optimizer (GWO) [5] has already been applied to TCN tuning [7]. Single-objective search, however, optimizes accuracy alone, and in a per-patient regime the cost of the selected configuration is not a secondary concern.

Both questions are addressed here. The contributions are: a controlled architecture study, in which an initial proposed deeper–narrower TCN is compared against the reference configuration under identical data, splits and budget (Section IV-A); a comparison of single-objective GWO against multi-objective GWO (MOGWO) [6] for tuning that architecture, which yields a Pareto front over accuracy and training cost (Sections IV-B–IV-D); a like-for-like comparison against the twelve-method benchmark of [3] at its own reported horizon (Section IV-E); and an evaluation over all 48 MIT-BIH records rather than the curated 21-record subset (Section IV-F).

II. RELATED WORK

Deep learning on ECG has been dominated by classification, from convolutional and recurrent classifiers to hybrids combining convolution, recurrence and attention with metaheuristic tuning [9], and to reservoir-computing implementations

under tight hardware budgets [10]. Forecasting is comparatively sparse: the waveform is predicted with LSTM networks in [11], while MIT-BIH records are treated as chaotic series and benchmarked against twelve predictors in [3], a framing supported there by positive largest Lyapunov exponents and connected to the wider chaotic time-series literature [12].

Convolutional sequence models were shown in [4] to match or exceed recurrent ones across standard benchmarks, using causal convolutions, exponentially increasing dilation, and residual connections. The design consequence relevant here is that the receptive field is fully determined by kernel size and depth, so a shallow network cannot be repaired by widening it.

The Grey Wolf Optimizer [5] models the leadership hierarchy and encircling behaviour of grey wolves; it requires no gradients, handles mixed discrete–continuous spaces, and has few control parameters. It was used in [7] to tune a TCN for transient stability assessment. The multi-objective extension MOGWO [6] maintains an external archive of non-dominated solutions and selects leaders from the least crowded archive segments, returning a Pareto front instead of a single point. That property is exploited here: in a per-record training regime, accuracy and training cost genuinely compete, and the practitioner rather than the optimizer should choose the trade-off. The MEALPY implementation [14] is used for the single-objective search.

Finally, ECG morphology and amplitude vary strongly between individuals, which is why inter-patient evaluation is much harder than intra-patient evaluation [13], and why methods such as Reversible Instance Normalization [8] have been proposed to absorb distribution shift. All results below are intra-patient; Section V bounds the claim accordingly.

III. METHODOLOGY

A. Dataset and protocol

The MIT-BIH Arrhythmia Database [15] provides 48 half-hour two-channel ambulatory recordings sampled at 360 Hz. The first 100,000 samples of the MLII lead of each record are split chronologically into 40,000 training, 10,000 validation and 50,000 test samples, following the protocol of [3] so that results remain directly comparable.

Two record sets are used and kept distinct. REF21 denotes the 21 records used in [3] (100–109, 112–117, 119, 121–124; records 111 and 118 are excluded there because the models failed to train on them), and every head-to-head comparison against that study uses it. ALL48 denotes the complete database, used to test whether the tuned configuration generalizes beyond the curated subset.

Each record is scaled to $[0, 1]$ by a Min–Max transform fitted on its own training split only, so that no test statistic leaks into training. Metrics are computed on the scaled signal, matching [3]; because the scaling is per-record and monotone, R^2 is unaffected by it and RMSE and MAE remain comparable to the values reported there. Inputs are windows of $L=10$ consecutive samples (27.8 ms of history); the target is the next H samples. Horizons $H \in \{1, 10, 15, 20\}$ are evaluated,

corresponding to 2.8, 27.8, 41.7 and 55.6 ms ahead. One model is trained per record and per horizon.

B. Forecasting strategy

Three multi-step strategies were measured on six representative records. Recursive forecasting, which feeds each prediction back as input, proved not merely less accurate but unstable, reaching a mean RMSE of 831 at $H=50$ against 0.0899 for multi-output. Direct forecasting, which trains H independent models, cost $8.7\times$ the training time of multi-output (360.9 s against 41.7 s per record at $H=10$) for a slightly worse result (RMSE 0.0389 against 0.0375). Multi-output (MIMO) was therefore both the most accurate and the cheapest, and is used throughout.

C. TCN architecture

The network is a stack of n residual blocks. Block i applies two causal 1-D convolutions with f filters, kernel size k and dilation 2^i , each followed by batch normalization, ReLU and dropout; a 1×1 convolution adapts the identity path when channel counts differ. The representation at the last time step passes through a dense layer of f units with ReLU and a linear layer of size H .

Because each block contributes two dilated convolutions, the receptive field is

$$R = 1 + 2(k-1) \sum_{i=0}^{n-1} 2^i = 1 + 2(k-1)(2^n - 1). \quad (1)$$

With $k=3$ this gives $R=5$ for $n=1$, $R=29$ for $n=3$ and $R=61$ for $n=4$. The first is smaller than the 10-sample input window: a single-block TCN physically cannot integrate the whole window through its convolutions. That observation motivates the architecture study of Section IV-A.

D. Configurations under comparison

Table I lists the four configurations. All share $k=3$, MSE loss, the Adam optimizer, 200 epochs, early stopping with patience 50, and learning-rate reduction on plateau.

TABLE I: TCN configurations. RF is the receptive field from (1); parameter counts are for $H=10$.

Parameter	Reference	Initial prop.	GWO	MOGWO
Residual blocks n	1	3	4	3
Filters f	256	64	256	128
Dropout d	0.100	0.100	0.109	0.131
Learning rate η	0.001	0.001	0.00207	0.00451
Batch size	128	256	128	128
Receptive field R	5	29	61	29
Parameters	266,497	67,905	1,456,138	268,042
Selected by	[3]	manual	GWO	MOGWO

E. Single-objective search: GWO

GWO [5] maintains a population of candidates; the three best (*alpha*, *beta*, *delta*) guide the rest through an encircling update whose exploration radius is governed by a coefficient decreasing linearly from 2 to 0. Three hyperparameters were searched with the rest held fixed ($k=3$, batch size 128): filter count $f \in \{32, 64, 128, 256\}$ encoded as an integer index, dropout $d \in [0.05, 0.40]$, and learning rate $\eta \in [10^{-4}, 10^{-2}]$, both continuous.

Evaluating a candidate means training a TCN on a five-record pilot set at $H=10$ with a reduced budget (100 epochs, patience 20) and returning the mean test RMSE,

$$\mathcal{F}(\theta) = \frac{1}{|P|} \sum_{p \in P} \text{RMSE}_p(\theta), \quad (2)$$

with P the pilot records. The search used 8 wolves over 4 iterations (≈ 32 evaluations). Its best solution, found at evaluation 5 with pilot RMSE 0.02615, was $f=256$, $d=0.109$, $\eta=0.00207$ at $n=4$, which defines the GWO configuration.

Sensitivity to the pilot set was probed by repeating the search with one record replaced by a harder one and with $n=3$. It converged to $f=128$, $d \approx 0.06$, $\eta \approx 0.0023$, the same filter region MOGWO selects, at a uniformly higher pilot RMSE (0.0306 against 0.0262) reflecting the harder pilot. Filter count and learning-rate scale are therefore stable under pilot perturbation; the absolute fitness value is not.

F. Multi-objective search: MOGWO

Objective (2) is blind to cost, and it duly selected the largest architecture in the space. Since a model is retrained per record, this is expensive. MOGWO [6] replaces the scalar fitness with a vector,

$$\min_{\theta} [\mathcal{F}_1(\theta), \mathcal{F}_2(\theta)] = [\overline{\text{RMSE}}(\theta), \overline{T}_{\text{train}}(\theta)], \quad (3)$$

and maintains an external archive of non-dominated solutions: θ_a dominates θ_b when it is no worse on both objectives and strictly better on at least one. Leaders are drawn from the archive by a roulette wheel biased toward sparsely populated regions, which spreads the front rather than collapsing it onto one point.

Depth was fixed at $n=3$ for this search on the grounds of (1): $R=29$ already covers the 10-sample window nearly three times over, so the extra block of the GWO configuration ($R=61$) buys receptive field that the input cannot supply. The search used 5 wolves over 5 iterations (30 evaluations) on the same pilot protocol, and returned a 7-solution archive.

G. Evaluation

RMSE, MAE and R^2 are reported over the flattened multi-step predictions of each record's test split, together with wall-clock training time per record (NVIDIA Tesla T4, TensorFlow 2.20). Because per-record errors are paired and not normally distributed, configuration comparisons use the two-sided Wilcoxon signed-rank test over records, and the number of records on which each configuration wins is reported alongside the mean, since the two can disagree.

IV. EXPERIMENTAL RESULTS

A. Receptive field versus width

Table II and Fig. 1 compare the reference configuration with the initial proposed architecture on REF21 across all four horizons, holding everything but the architecture fixed.

TABLE II: Reference configuration against the initial proposed architecture, mean over REF21. p is the Wilcoxon signed-rank test on paired per-record RMSE.

H	Config	RMSE	MAE	R^2	Time (s)	p
1	Reference	0.0061	0.0037	0.9952	226.6	0.203
	Initial prop.	0.0052	0.0036	0.9970	169.8	
10	Reference	0.0344	0.0148	0.8776	207.7	$5 \cdot 10^{-5}$
	Initial prop.	0.0304	0.0134	0.8991	159.5	
15	Reference	0.0471	0.0200	0.7847	202.0	$2 \cdot 10^{-4}$
	Initial prop.	0.0433	0.0185	0.8108	160.4	
20	Reference	0.0571	0.0253	0.6894	188.6	$3 \cdot 10^{-3}$
	Initial prop.	0.0539	0.0236	0.7157	156.4	

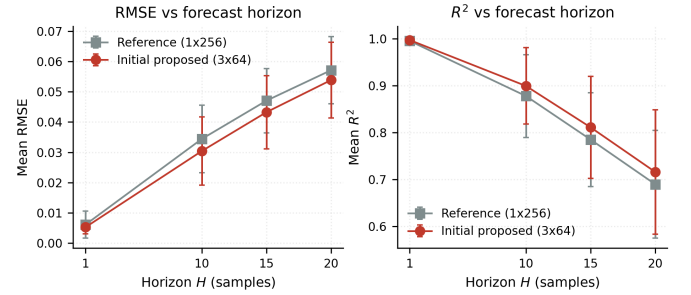
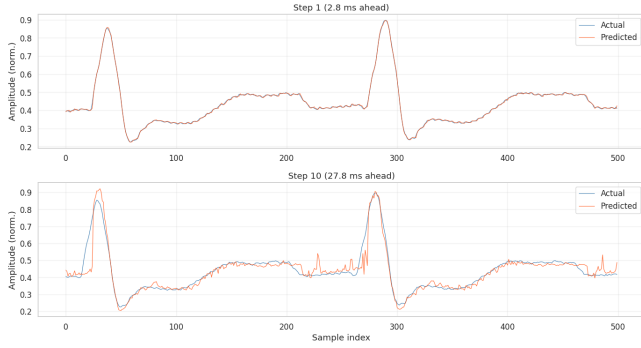


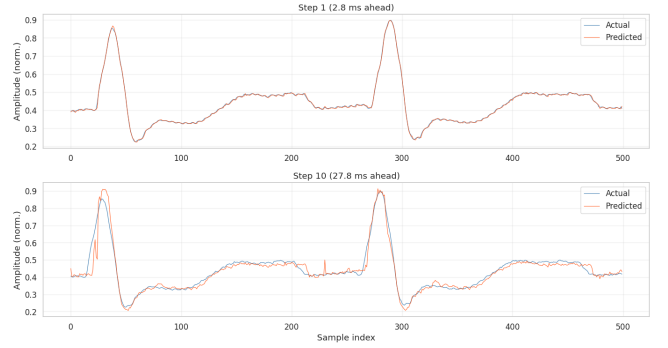
Fig. 1: Mean RMSE and R^2 against forecast horizon over REF21. Error bars are one standard deviation across records.

The initial proposed architecture is better on every metric at every horizon, and cheaper: 17–25% less training time with $3.9\times$ fewer parameters. The gain is significant at $H=10$, 15 and 20 (19–20 record wins out of 21), but not at $H=1$ ($p=0.203$, 12/21 wins); at a single step the task is easy enough that both configurations sit near ceiling ($R^2 > 0.995$) and the architectural difference does not separate them reliably. This is stated explicitly because the mean difference at $H=1$, 14.1%, is the largest of the four in relative terms and would otherwise invite over-reading.

The mechanism is receptive-field coverage. The reference stack sees 5 of the 10 input samples by (1); the rest reaches the output only through the dense head, so widening to 256 filters adds capacity but no context. Three blocks at 64 filters give $R=29$, and the advantage grows with the horizon in absolute R^2 terms, from +0.0018 at $H=1$ to +0.0263 at $H=20$, as expected if longer context is what longer-horizon prediction needs. Fig. 2 shows the difference on record 109: at step 1 both configurations track the waveform closely, while at step 10 the reference configuration produces visibly noisier excursions along the ST segment and overshoots the R peak more sharply.



(a) Reference configuration [3].



(b) Initial proposed architecture.

Fig. 2: Record 109 at $H=10$: predicted against actual waveform at the first and last forecast step. The step-1 traces are nearly indistinguishable; at step 10 the reference configuration shows larger baseline noise and a sharper R-peak overshoot.

B. Single-objective GWO

The GWO-selected configuration was retrained on all REF21 records at $H=10$ with the full budget. It improves on the initial proposed architecture by 2.4% RMSE and 3.8% MAE (Table III), winning on 20 of 21 records ($p=4\times 10^{-4}$), and lifts mean R^2 above 0.90.

It also costs $3.7\times$ more training time per record and holds $21\times$ more parameters. The search behaved exactly as its objective instructed: with fitness defined as RMSE alone, the largest filter count in the space is never penalized, and GWO converged onto it at evaluation 5 of 32, after which the remaining 27 evaluations bought nothing. This is the failure mode multi-objective search is meant to correct.

TABLE III: Optimized configurations against the hand-set baseline at $H=10$, mean over REF21. p from Wilcoxon signed-rank on per-record RMSE against the initial proposed architecture.

Config	RMSE	MAE	R^2	Time (s)	Params
Reference	0.03441	0.01484	0.8776	207.7	266,497
Initial prop.	0.03042	0.01340	0.8991	159.5	67,905
GWO	0.02970	0.01289	0.9020	591.5	1,456,138
MOGWO	0.02967	0.01299	0.9040	273.9	268,042

GWO vs. initial prop.: -2.36% RMSE, 20/21 wins, $p=4\times 10^{-4}$
MOGWO vs. initial prop.: -2.47% RMSE, 14/21 wins, $p=0.055$
MOGWO vs. GWO: -0.11% RMSE, 4/21 wins, $p=0.029$

C. Multi-objective GWO and the Pareto front

Fig. 3 shows the 30 MOGWO evaluations in objective space and the 7-solution non-dominated archive, which spans 127–163 s of pilot training time and 0.0261–0.0300 pilot RMSE. Its structure is informative: the 256-filter candidates sit far to the right, 223–235 s, without reaching a lower RMSE than the best 128-filter candidates, so they are dominated and never enter the archive; 32 filters is dominated from the other side, being both slow to converge and inaccurate. Six of the seven archive members use 128 filters, and dropout is the parameter that trades along the front.

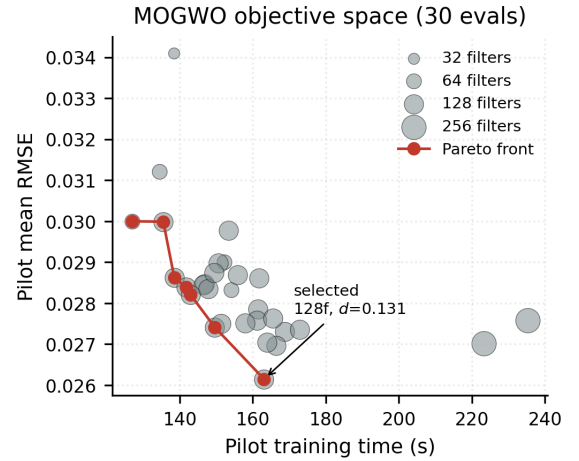


Fig. 3: MOGWO objective space over 30 evaluations. Marker size encodes filter count; the highlighted curve is the non-dominated archive.

The lowest-RMSE archive member was selected ($f=128$, $d=0.131$, $\eta=0.00451$; pilot RMSE 0.026148 against GWO’s 0.026152) and evaluated under the full budget. On REF21 at $H=10$ it reaches RMSE 0.02967 and $R^2=0.9040$, a mean marginally better than the single-objective result, obtained with $5.4\times$ fewer parameters and $2.2\times$ less training time per record.

The per-record picture is more nuanced than the mean. The larger GWO model wins on 17 of the 21 records, and the Wilcoxon test detects this ($p=0.029$); the MOGWO mean advantage comes from the tail, principally record 101, the hardest in the set, where the 1.46M-parameter model overfits (RMSE 0.0639, $R^2=0.662$) while the 268K-parameter model does not (0.0520, $R^2=0.777$). The two are therefore equivalent in aggregate accuracy, the GWO configuration slightly more accurate on typical records and the MOGWO configuration markedly more robust on difficult ones at a fraction of the training cost.

D. The accuracy–cost frontier

Fig. 4 places the four configurations in the accuracy–size and accuracy–time planes. The reference configuration is Pareto-dominated outright, being simultaneously the least accurate and the second largest. Scaling from 68K to 1.46M parameters, a factor of 21, buys 2.4% RMSE, so the accuracy–capacity curve for this task is essentially flat past a few hundred thousand parameters. The MOGWO configuration sits at the knee, matching the largest model’s accuracy at 18% of its size.

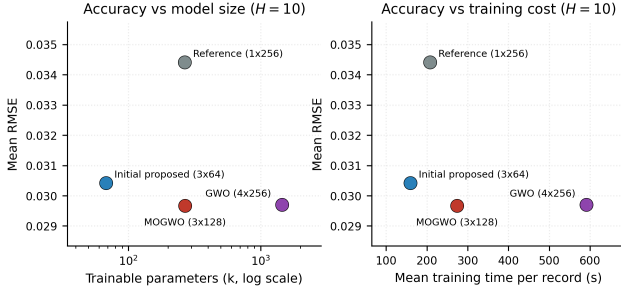


Fig. 4: Accuracy against model size (left) and against per-record training time (right) at $H=10$ over REF21.

E. Single-step benchmark against the reference study

Single-step-ahead prediction for twelve architectures on REF21 is reported in [3]. Because the protocol here reproduces that one, with the same records, chronological split, input window, training budget and per-record scaling, the published values can be compared directly with the $H=1$ results obtained here on the same records. The comparison is shown in Fig. 5; the reference values are those tabulated in [3] and are not reproduced here.

The MOGWO-tuned TCN attains a mean RMSE of 0.00499 against 0.0082 for TCN–LSTM, the strongest of the twelve, a 39.1% reduction, with MAE 0.00352 against 0.0051 and R^2 0.99726 against 0.9908. It wins on 20 of the 21 records (Wilcoxon $p=2.4 \times 10^{-5}$); the single loss is record 104, a noisy recording with pacing artefacts, where the RMSE of 0.00969 exceeds the 0.0073 reported in [3].

One caveat deserves emphasis. The reproduction of the reference configuration itself also outperforms the published TCN figure, 0.0061 against 0.0085 RMSE. Since architecture, split, window and budget match, the residual difference is attributable to the training stack rather than to the model: current TensorFlow, Adam with plateau-based learning-rate reduction, batch normalization inside every residual block, and 2019-era against current GPU numerics. The conclusion is therefore twofold: against a like-for-like modern reimplement, tuning contributes the step from 0.0061 to 0.00499, an 18% reduction, while against the published numbers the total is 39.1%, part implementation and part architecture and tuning. Both are reported rather than the larger figure alone.

F. Scaling to the full database

REF21 excludes 27 records, among them records 111 and 118, dropped in [3] because the models there produced outliers on them, and the entire 200-series, which contains the more severe arrhythmias. Retraining the MOGWO configuration on all 48 records gives mean RMSE 0.00530 ± 0.00204 and R^2 0.9970 ± 0.0026 at $H=1$, and mean RMSE 0.03307 ± 0.01277 and R^2 0.8865 ± 0.0794 at $H=10$.

Single-step performance is essentially unchanged by the wider record set, and records 111 and 118 train without difficulty here (R^2 0.9957 and 0.9982). At $H=10$ the spread across records is large: the 200-series records dominate the lower tail (209: $R^2=0.693$; 215: 0.687; 210: 0.715), while the best remain above 0.97. Multi-step forecasting is thus far more sensitive to arrhythmia burden than single-step forecasting, since an ectopic beat is precisely an event the recent past does not predict. Per-step decomposition shows RMSE growing close to linearly from roughly 0.008 at step 1 to 0.053 at step 10, with no sign of a fallback to predicting a constant; as Fig. 2 illustrates, the dominant error mode at longer horizons is timing rather than amplitude.

V. DISCUSSION AND LIMITATIONS

Three limitations bound what has been shown. First, the optimization was performed at one horizon: both searches used $H=10$ pilots, and the selected configurations were then applied unchanged elsewhere. The strength of the $H=1$ results suggests the configuration transfers, but this was not optimized for and is not established. Second, depth was fixed rather than searched, at 4 blocks in the single-objective run and 3 in the multi-objective one, on the receptive-field argument of (1). That argument is sound but is an argument, not a measurement, and the two searches consequently differ in two respects rather than one. Third, the evaluation is intra-patient. A model per record needs 40,000 labelled samples from that record before it is useful, which is a real deployment constraint.

A preliminary experiment with Reversible Instance Normalization [8] shows how sharply the regime matters. Trained on 15 records and tested on 6 unseen ones it improved RMSE by 6.8%, the behaviour the method predicts; applied on top of per-record scaling in the intra-patient regime used here it degraded RMSE by 17.7% and improved on only 3 of 21 records ($p=0.0038$), since once training and test data share a record and a fitted scaler the per-window statistics it removes are signal rather than nuisance.

On validity: metrics are computed on the Min–Max-scaled signal to match [3], so RMSE and MAE carry the scale of each record’s training-split range, which is why they are paired with R^2 and with per-record paired tests rather than compared across records; and the comparison in Section IV-E inherits whatever implementation differences separate the two training stacks, which is why the reproduction is reported alongside.

VI. CONCLUSION AND FUTURE WORK

TCN architectures and their automated tuning were studied for multi-step ECG forecasting on MIT-BIH under a controlled

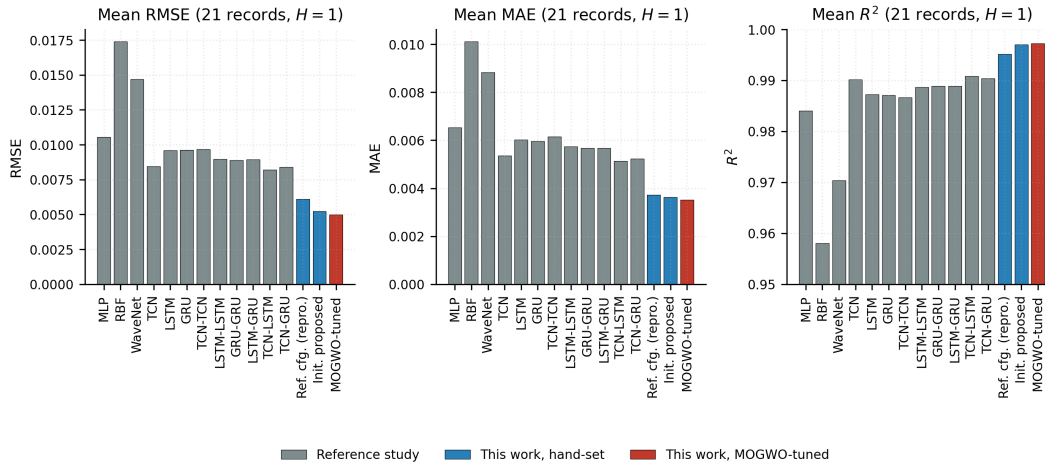


Fig. 5: Single-step ($H=1$) performance on REF21: the twelve architectures benchmarked in [3] (grey), the reproduction of their configuration and the initial proposed architecture (blue), and the MOGWO-tuned TCN (red). Lower is better for RMSE and MAE, higher for R^2 . Note the truncated R^2 axis.

protocol. Receptive-field coverage, not width, drives accuracy: an initial proposed three-block, 64-filter TCN beats the wide-shallow reference configuration at every horizon with $3.9\times$ fewer parameters and 17–25% less training time. Single-objective GWO improves accuracy by a further 2.4% but selects a 1.46M-parameter model, since its objective never sees the cost; MOGWO returns a Pareto front whose knee matches that accuracy with $5.4\times$ fewer parameters and $2.2\times$ less training time, and is more robust on the hardest records. At the single-step horizon reported in [3], the tuned model reduces mean RMSE by 39.1% relative to the best of the twelve architectures benchmarked there, winning on 20 of 21 records, roughly half of that margin being attributable to the modern training stack.

Four directions follow. Depth and kernel size should be included in the MOGWO search space, so that the front is traced over the architecture rather than around a fixed skeleton. The objective should be defined across horizons rather than at $H=10$ alone. Inference latency and parameter count should be added as explicit objectives, since these are what an embedded deployment actually constrains. And the inter-patient gap should be attacked directly: the intra-patient problem is close to solved at $H=1$, and the remaining headroom lies in generalizing to patients whose data the model has never seen.

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